

Problem 21 (USA Math Olympiad Program). Let p be an odd prime. Find the number of 6-tuples (a, b, c, d, e, f) of integers between 0 and $p - 1$ such that

$$a^2 + b^2 + c^2 \equiv d^2 + e^2 + f^2 \pmod{p}.$$

Solution. Let $\zeta = e^{\frac{2\pi i}{p}}$. Using the roots of unity filter, the number N of such 6-tuples is

$$\begin{aligned} \frac{1}{p} \sum_{a,b,c,d,e,f \in \mathbb{F}_p} \sum_{j=0}^{p-1} \zeta^{j(a^2+b^2+c^2-d^2-e^2-f^2)} &= \frac{1}{p} \sum_{j=0}^{p-1} \sum_{a,b,c,d,e,f \in \mathbb{F}_p} \zeta^{j(a^2+b^2+c^2-d^2-e^2-f^2)} \\ &= \frac{1}{p} \sum_{j=0}^{p-1} G(j, p)^3 G(-j, p)^3, \end{aligned}$$

where $G(m, p) = \sum_{a \in \mathbb{F}_p} \zeta^{ma^2}$ is the quadratic Gauss sum. Note that for $(m, p) = 1$, we have $G(m, p) = \left(\frac{m}{p}\right) G(1, p)$ and $G(1, p)^2 = \left(\frac{-1}{p}\right) p$ where $\left(\frac{m}{p}\right)$, etc. denotes the Legendre symbol. This gives

$$\begin{aligned} N &= \frac{1}{p} \left(p^6 + \sum_{j=1}^{p-1} \left(\frac{j}{p}\right)^3 G(1, p)^3 \left(\frac{-j}{p}\right)^3 G(1, p)^3 \right) \\ &= \frac{1}{p} \left(p^6 + \sum_{j=1}^{p-1} \left(\frac{j}{p}\right)^6 \left(\frac{-1}{p}\right)^6 p^3 \right) \\ &= \frac{1}{p} (p^6 + (p-1)p^3) = p^5 + p^3 - p^2. \end{aligned}$$

□

Problem 22 (IMO Shortlist 2007 C3). Find all positive integers n for which the numbers in the set $S = \{1, 2, \dots, n\}$ can be coloured red and blue, with the following condition being satisfied: The set $S \times S \times S$ contains exactly 2007 ordered triples (x, y, z) such that:

- (i) The numbers x, y, z are of the same colour and
- (ii) The number $x + y + z$ is divisible by n .

Solution. The answer is $n \in \{69, 84\}$.

Let $\zeta = e^{\frac{2\pi i}{n}}$. Let B and R denote the set of numbers coloured blue and red, respectively, and define

$$B_j = \sum_{k \in B} \zeta^{jk}, \quad R_j = \sum_{k \in R} \zeta^{jk}.$$

Clearly,

$$B_j + R_j = \begin{cases} n & \text{if } j = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Let N denote the number of triples satisfying both conditions. Applying the roots of unity filter, we have

$$\begin{aligned} N &= \frac{1}{n} \sum_{j=0}^{n-1} \left(\sum_{x,y,z \in B} \zeta^{j(x+y+z)} + \sum_{x,y,z \in R} \zeta^{j(x+y+z)} \right) = \frac{1}{n} \sum_{j=0}^{n-1} (B_j^3 + R_j^3) \\ &= \frac{1}{n} \sum_{j=0}^{n-1} (B_j + R_j)((B_j + R_j)^2 - 3B_j R_j) = \frac{1}{n} (B_0 + R_0)((B_0 + R_0)^2 - 3B_0 R_0) = n^2 - 3B_0 R_0. \end{aligned}$$

For brevity, let $B_0 = m$, which is the number of elements in blue. Then we wish to find all integers n such that the equation

$$n^2 - 3m(n - m) = n^2 - 3mn + 3m^2 = 2007$$

has a solution with $m \leq n$. Thus the discriminant is a perfect square, so

$$4 \cdot 2007 - 3m^2 = t^2$$

for some integer t . By taking mod 3, it follows that m and t are divisible by 3, so there are some integers x, y such that

$$x^2 + 3y^2 = 4 \cdot 223 = 892.$$

This bounds $y \leq 17$. An exhaustive search shows that the only possible solutions (x, y) are $(28, 6), (23, 11), (5, 17)$. Given that $m \leq n$, the solutions are $(m, n) = (18, 69), (33, 84), (51, 84), (51, 69)$. This means $n = 84$ or 69 . $\square$

Problem 23 (American Mathematical Monthly). Let p be an odd prime. Prove that the $2^{\frac{p-1}{2}}$ numbers $\pm 1 \pm 2 \pm \dots \pm \frac{p-1}{2}$ represent each nonzero residue class mod p the same number of times. What is this number?

Solution. Let $\zeta = e^{\frac{2\pi i}{p}}$. Applying the roots of unity filter, the number N_r of expressions congruent to $r \pmod{p}$ is

$$\frac{1}{p} \sum_{j=0}^{p-1} \sum_{a_k = \pm 1} \zeta^{j(a_1 + 2a_2 + \dots + \frac{p-1}{2}a_{\frac{p-1}{2}} - r)} = \frac{1}{p} \sum_{j=0}^{p-1} \zeta^{-jr} \prod_{k=1}^{\frac{p-1}{2}} (\zeta^{jk} + \zeta^{-jk}).$$

For $j = 0$, the summand is $2^{\frac{p-1}{2}}$. For $j \neq 0$, the two expressions $\zeta^{jk_1} + \zeta^{-jk_1}$ and $\zeta^{jk_2} + \zeta^{-jk_2}$ are equal iff $k_1 \equiv \pm k_2 \pmod{p}$. So the terms in the product form a permutation of $\zeta + \zeta^{-1}, \zeta^2 + \zeta^{-2}, \dots, \zeta^{\frac{p-1}{2}} + \zeta^{-\frac{p-1}{2}}$. So we have

$$N_r = \frac{1}{p} \left(2^{\frac{p-1}{2}} + \sum_{j=1}^{p-1} \zeta^{-jr} \prod_{k=1}^{\frac{p-1}{2}} (\zeta^k + \zeta^{-k}) \right) = \frac{1}{p} \left(2^{\frac{p-1}{2}} - \prod_{k=1}^{\frac{p-1}{2}} (\zeta^k + \zeta^{-k}) \right)$$

for $r \neq 0$. In particular, this shows N_r is a constant for any $r \neq 0$.

Let $S = \prod_{k=1}^{\frac{p-1}{2}} (\zeta^k + \zeta^{-k})$. With some algebraic manipulation,

$$\prod_{k=1}^{p-1} (\zeta^k + \zeta^{-k}) = S \cdot \prod_{k=\frac{p+1}{2}}^{p-1} (\zeta^{k-p} + \zeta^{p-k}) = S^2.$$

On the other hand,

$$\prod_{k=1}^{p-1} (\zeta^k + \zeta^{-k}) = \zeta^{\frac{-p(p-1)}{2}} \prod_{k=1}^{p-1} (1 + \zeta^{2k}) = \prod_{k=1}^{p-1} (1 + \zeta^k).$$

The last equality comes from the fact that $\zeta^k \mapsto \zeta^{2k}$ is a permutation of the primitive roots of unity. By considering the roots, we have the identity $x^p - 1 = (x - 1)(x - \zeta)(x - \zeta^2) \dots (x - \zeta^{p-1})$. So

$$\prod_{k=1}^{p-1} (1 + \zeta^k) = \prod_{k=1}^{p-1} (-1 - \zeta^k) = \frac{(-1)^p - 1}{(-1) - 1} = 1,$$

and hence $S = \pm 1$. It follows that

$$N_r = \frac{1}{p} (2^{\frac{p-1}{2}} \pm 1).$$

Since this is an integer, and $2^{\frac{p-1}{2}} \equiv \left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}} \pmod{p}$, we have

$$N_r = \frac{2^{\frac{p-1}{2}} - (-1)^{\frac{p^2-1}{8}}}{p}.$$

$\square$

Problem 24 (Taiwan TST 2019). Given a prime $p = 8k + 1$ for some integer k , let r be the remainder when $\binom{4k}{k}$ is divided by p . Prove that $\sqrt{r}$ is not an integer.

Solution. We do everything in $\mathbb{F}_p$. Let g be a primitive root and $\zeta = g^4$ (i.e. ζ is a $(2k)$ th root of unity.) Then

$$\frac{1}{2k} \sum_{j=0}^{2k-1} (1 + \zeta^j)^{4k} = \binom{4k}{0} + \binom{4k}{2k} + \binom{4k}{4k}.$$

On the other hand, ζ^2 is a k th root of unity, so

$$\frac{1}{k} \sum_{j=0}^{k-1} (1 + \zeta^{2j})^{4k} = \binom{4k}{0} + \binom{4k}{k} + \binom{4k}{2k} + \binom{4k}{3k} + \binom{4k}{4k}.$$

Hence

$$\begin{aligned} 4k \binom{4k}{k} &= 2 \sum_{j=0}^{k-1} (1 + \zeta^{2j})^{4k} - \frac{1}{2k} \sum_{j=0}^{2k-1} (1 + \zeta^j)^{4k} \\ &= 2 \sum_{\substack{j=0 \\ 2 \nmid j}}^{2k-1} (1 + \zeta^j)^{4k} - \sum_{j=0}^{2k-1} (1 + \zeta^j)^{4k} \\ &= \sum_{\substack{j=0 \\ 2 \nmid j}}^{2k-1} (1 + \zeta^j)^{4k} - \sum_{\substack{j=0 \\ 2 \nmid j}}^{2k-1} (1 + \zeta^j)^{4k}. \end{aligned}$$

Among the $2k$ terms on the right, only the term $(1 + \zeta^k)^{4k}$ is 0, and all other terms are ± 1 . Let n be the number of terms that are 1. Then the right-hand side is $n(1) + (2k - 1 - n)(-1) = 2n - 2k + 1$. Multiplying both sides by 2, and noticing $8k = -1$ in $\mathbb{F}_p$, we get

$$r = -2(2n - 2k + 1) \in \{4k - 2, 4k - 6, \dots, 2, -2, \dots, -4k + 2\}.$$

Since $0 \leq r \leq 8k$, we must have $r = 4k - 2, 4k - 6, \dots, 2, 8k - 1, 8k - 5, \dots, 4k + 3$. Each of these is congruent to 2 or 3 modulo 4, so r is not a perfect square.

(Note: When $k = 9$, $p = 73$, $\binom{36}{9} \equiv 71 \equiv 12^2 \pmod{73}$ is still a quadratic residue.) □